

Short-Text Classification Based on ICA and LSA

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Abstract. Many applications, such as word-sense disambiguation and information retrieval, can benefit from text classification. Text classifiers based on Independent Component Analysis (ICA) try to make the most of the independent components of text documents and give in many cases good classification effects. Short-text documents, however, usually have little overlap in their feature terms and, in this case, ICA can not work well. Our aim is to solve the short-text problem in text classification by using Latent Semantic Analysis (LSA) as a data preprocessing method, then employing ICA for the preprocessed data. The experiment shows that using ICA and LSA together rather than only using ICA in Chinese short-text classification can provide better classification effects.

1 Introduction

Text classification (TC) according to documents' topics is an application of automatic analysis of documents in modern information processing systems. ICA applied on word context data can give distinct features which reflect syntactic and semantic categories [1]. The application of ICA in TC is based on the assumption that a document corpus is generated by a combination of several thematic topics. Thus, the independent components obtained by the ICA algorithms define statistically independent clusters of documents, allowing their thematic classification [2]. Text classifiers based on ICA, in many cases, give good classification effects. In short-text case, however, ICA can not work well because short-text documents usually have little overlap in their feature terms.

Our aim is to solve the short-text problem we called STP in text classification by using LSA as a data preprocessing method. ICA is then employed for the preprocessed data by utilizing its great classifying capability of identifying a generalizable low-dimensional basis set in the face of high-dimensional noisy data [3].

2 LSA and ICA Related to TC

TC methods are almost depended on document vector space model (VSM) [4] representation that our analyses and experiments are also based on.

Latent semantic analysis (LSA) [5] is a well-known simple approach to automatic generation of the concepts and analysis of terms co-occurrence that are useful in text classification and information retrieval purposes. LSA is based on singular value

decomposition (SVD) [6] by which a smaller document matrix A that retains only the most important information from the original document matrix can be constructed.

$$A = TSD^T. \quad (1)$$

In (1), T and D contain orthogonal vectors and S is diagonal matrix of singular values. The eigenvectors with the largest singular value capture the axes of largest variation in data. So each document can be projected into a lower dimensional space, which is called dimensionality reduction often used to reduce data noise, avoid overlearning. After getting the reduced space, same co-occurrence terms are projected on the same dimensionality and documents are represented as linear combinations of orthogonal features. [4], [6] have introduced the applications of LSA in TC. [5] describes LSA in detail.

ICA is a method for presenting a set of multivariate observations as a linear combination of unknown latent variables that are statistically independent, giving no information on the values of the other components [7]. Lately it has been found out that it is a suitable method for analyzing text documents [3], [8].

We suppose that there are some main topics acting as a set of independent hidden random variables in each document. ICA as a technique that exploits the more effective higher order statistical features in documents can be employed to determine these independent random variables and extract the independent topic structure of documents. The classic ICA model can be expressed as following:

$$x = As. \quad (2)$$

In (2), x is the documents topic-mixed information, a $T \times D$ term by document matrix. A is a $T \times N$ mixing matrix and S is a $N \times D$ documents topic information matrix, N is the number of independent components representing main topics of a set of documents. If mixing matrix A is known, topics in a document corpus can be estimated as $y = A^{-1}x = Wx$, where separating matrix W can be determined by employing FastICA [7] algorithm. Consequently, new unseen documents can be classified under N topic categories by comparing their independent components with the ones of training documents, after projecting them to the ICA space by means of matrix W [2].

3 Short-Text Classification Problem

Each document has its special topic used for TC. Topics are considered as a set of independent hidden random variables. In VSM, suitable feature terms consisting of document vectors often implicate underlying topic of documents. Likewise, a topic can also determine which terms to appear in documents. If two documents, usually long-text documents, have some common feature terms, the same topic the two documents probably belong to. If regarding a set of documents as a topic-mixed corpus, TC based on ICA is then an application that ICA algorithm can maximize the statistical independence between documents from the observed topic-mixed document corpus and try to determine the documents' real topic categories.

But short-text documents usually have little overlap in their feature terms, especially after filtering stop-words or function-words. Fig.1 illustrates this case. This means that vectors of such documents are already independent of each other so that

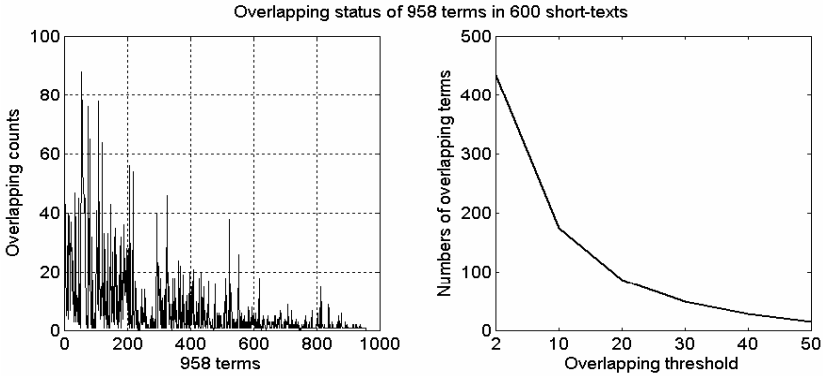


Fig. 1. Terms overlap in short-text documents. There are 958 terms in our 600 short-text documents including 6 topic categories. Left figure shows terms are in their overlap case and only a few terms are overlap terms. Right one clearly shows there are total 524 single terms (hapax legomena) if term overlap threshold is set greater than 2. In order to get an improved classification precision, the overlap threshold will be a larger number, e.g. 20. In this case, experiment shows only 87 terms among 958 are overlapped, that is, only 9.1% terms are overlapped.

ICA leads to nothing. Experiments in section 4 will show an unsatisfied classifying effect when only using ICA on Chinese short-text documents classification.

Fortunately, there is a possible solution to STP problem. Low terms overlap proportion is insufficient to determine that there is no similarity between documents because the co-occurrence terms may exist among different documents. LSA is just the right tool to find these co-occurrence terms by projecting them on the same dimensionality in reduced space. Co-occurrence terms can improve the similarity of short-text documents, that is, a better topic-mixed corpus is provided for ICA to extract the independent topic structure of documents.

4 Experiments

We randomly select 400 Chinese short-text documents downloaded from Chinese web sites (40930 terms after web-page preprocessing, Chinese terms splitting and out-of-vocabulary words recognizing by means of our programs) manually divided into 4 topic categories, such as *traffic* (TRA), *education* (EDU), *medicine* (MED) and *politics* (POL). Each topic contains 100 short documents.

Stop-words and function-words are removed. Jarque-Bera test shows that the remaining 660 terms are not normally distributed, so Chi-square [4] statistic is used to calculate these terms χ^2 -value over the four thematic topics. Experiments show that topic category of each document can be better represented by each document's 12 terms with higher χ^2 -value. The 12 terms are then selected as document feature terms, weighted by normalized $tf \times idf$ [4]. Each document is represented as 12-dimensional vector, $\mathbf{x}_j = (s_{1j}, \dots, s_{Kj})$, $K = 12$, where s_{ij} is:

$$s_{ij} = \text{round} \left(10 \times \frac{1 + \lg(tf_{ij})}{1 + \lg(l_j)} \right) \tag{3}$$

In (3), tf_{ij} represents the occurrence of term i in document j and l_j is the length of document j . The calculation of s_{ij} employs the logarithm method, considering standardization of weights factor.

As a data preprocessing step to solve STP, LSA is employed to reduce the original 660 dimensionalities to 17. The number of independent components is set to 4 representing our four selected topic categories. The classifier is trained with the 80% of the short-text documents (320 documents randomly selected) in corpus by FastICA algorithm. Classifier obtained from FastICA determines projection directions into which short-text document must be projected to reveal the latent topic that the document belong to. As a contrast, a classifier only based on ICA is trained in the same corpus.

We use the remaining 80 documents in training set (denoted as *In_Topics_Test*) and 200 documents from other two topic categories (denoted as *Out_Topics_Test*): *military* (MIL) and *sports* (SPO) as testing set to verify classification effect of the classifier obtained in training phase. Fig. 2 shows the classifying results of the 80 documents in testing set by using LSA and ICA together (denoted as *Method_1*) and ICA only (denoted as *Method_2*) respectively.

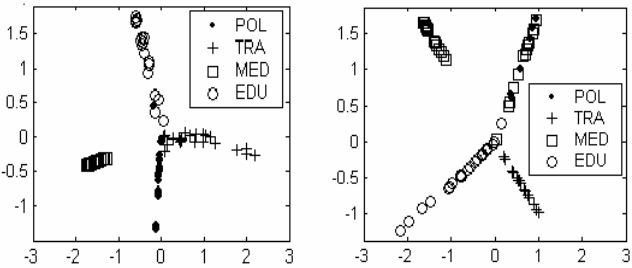


Fig. 2. The left figure is the classifying result obtained by *Method_1*, the right one shows the classifying result obtained by *Method_2*. In right figure, topics of *politics* and *medicine* are confused due to their little overlap feature terms. However, the left figure shows that four topic categories are clearly classified, proving that *Method_1* is suitable for short-text classification.

The detail classifying effects of Fig. 2 are illustrated in Table 1. According to the average within-class scatter (*Method_1*'s 0.99 vs. *Method_2*'s 3.75) and average between-class scatter (*Method_1*'s 2.35 vs. *Method_2*'s 1.23), *Method_1* in short-text classification makes a better classifying effect than that of *Method_2*. At the same time, *Method_1* gets a higher classifying precision than that of *Method_2*, such as in *politics* topic, classifying precision is *Method_1*'s 95% vs. *Method_2*'s 30%.

There are two classifiers learnt by means of *Method_1* and *Method_2*. Fig. 3 and Fig. 4 show the classifying capability of two classifiers learnt in training phase. There are two kinds of lines (real line and broken line) we called *topic status curve* (TSC) in these figures. One TSC is represented by average value of independent component of each document in one topic category. Real lines represent TSCs in training set, and broken lines represent TSCs in testing set.

Table 1. Comparison of classifying effects between *Method_1* and *Method_2*. C denotes four categories (*traffic*, *politics*, *education*, *medicine*), and M is the two classifying method. WS shows ‘within-class scatter’ in each topic category respectively. Each three values in BS corresponds to ‘between-class scatter’, such as topic *traffic*, the values (2.76, 0.86, 2.69) correspond to the ‘between-class scatter’ of *traffic* with *politics*, *medicine*, *education*, respectively. N is the number of documents classified in the corresponding topic category.

C M	TRA			POL			MED			EDU		
	WS	BS	N	WS	BS	N	WS	BS	N	WS	BS	N
ICA	0.96	2.76	20	5.83	2.76	6	4.60	0.86	33	4.61	2.69	21
		0.86			0.54			0.54			0.003	
		2.69			0.003			0.51			0.51	
LSA +ICA	0.52	0.57	20	5.02	0.57	18	0.21	2.85	20	3.66	1.14	22
		2.85			0.87			0.87			0.10	
		1.14			0.10			0.39			0.39	

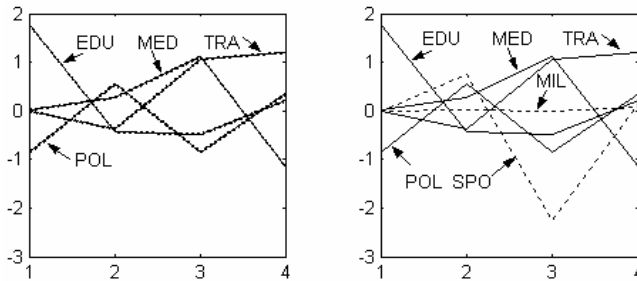


Fig. 3. The classifying effects when using classifier learnt by *Method_1*. Left figure shows the surprising similarity of the TSCs in training set and in *In_Topics_Test* set, the two kinds of TSCs are almost overlapped. That is, the classifying precision is perfect when documents are in *In_Topics_Test* set. In right figure, TSCs of *Out_Topics_Test* are very different from the four TSCs, which illustrates that the classifier learnt by *Method_1* can reject documents from *Out_Topics_Test* set and do not classify them into the four training topic categories.

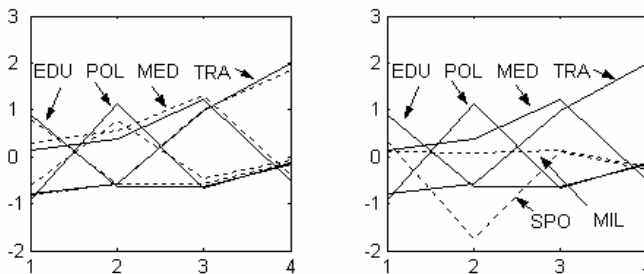


Fig. 4. The classifying effects when using classifier learnt by *Method_2*. Left figure shows the similarity of the TSCs in training set and in *In_Topics_Test* set. However, real lines and its corresponding broken line do not overlap well, that means, the classifying precision is unsatisfied when documents are in *In_Topics_Test* set. The rejection capability of this kind of classifier learnt by *Method_2* is almost as same as the classifier learnt by *Method_1*.

5 Conclusion

In this study, we have analyzed short-text problem in TC. This kind of problem can be solved without loss of generality by LSA projection. Experimental results illustrate that ICA and LSA together provide a satisfied classifying result in Chinese short-text classification, that is, LSA is good at analyzing terms co-occurrence, extracting and inferring relations of contextual usage of terms in documents, increasing the semantic terms overlapping, while ICA finds statistically independent topics, providing a great classifying capability. Using ICA only, however, can not obtain a satisfied classifying result. At the same time, experiments show that Chi-square statistic is a suitable method for extracting feature terms in short-text documents.

How many dimensionalities of the reduced dimensionality space should be determined when using LSA projection? In this study, the dimensionality of reduced space comes from experiment that makes ICA yield a good classifying effect. In future works, however, the dimensionality should be determined in manner of quantitative analysis and word-sense disambiguation will be taken into account in TC.

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